Properties of addition and multiplication

Commutative properties

$$A + B = B + A$$

$$AB = BA$$

Associative properties

$$(A + B) + C = A + (B + C)$$

$$(AB)C = A(BC)$$

Distributive property

$$A(B + C) = AB + AC$$

Working with negatives

Definition of subtaction

$$A - B = A + (-B)$$

Properties of negatives

$$A + (-A) = 0$$

$$-(-A) = A$$

$$-(A + B) = -A + (-B)$$

$$-(AB) = (-A)B = A(-B)$$

$$(-A)(-B) = AB$$

$$-\frac{A}{B} = \frac{-A}{B} = \frac{A}{-B}$$

$$\frac{-A}{-B} = \frac{A}{B}$$

Working with fractions

Definition of division

$$A \div B = \frac{A}{B}$$

Properties of fractions

$$A \cdot \frac{1}{A} = 1$$

$$\frac{1}{\frac{1}{A}} = A$$

$$\frac{A}{1} = A$$

$$\frac{A}{C} + \frac{B}{C} = \frac{A+B}{C}$$

$$\frac{A}{B} \cdot \frac{C}{D} = \frac{AC}{BD}$$

$$A \cdot \frac{B}{C} = \frac{AB}{C}$$

$$\frac{A}{B} = \frac{C}{D} \iff \frac{B}{A} = \frac{D}{C} \iff AD = BC$$

Properties of lines

The line with slope m and y-intercept b has equation $y = mx + b$.

The line with slope m passing through the point $(x_0,\ y_0)$ has equation $y - y_0 = m(x - x_0)$.

The line through points $(x_1,\ y_1)$ and $(x_2,\ y_2)$ has slope $\dfrac{y_2 - y_1}{x_2 - x_1}$.

The line segment with endpoints $(x_1,\ y_1)$ and $(x_2,\ y_2)$ has midpoint $\left(\dfrac{x_1 + x_2}{2},\ \dfrac{y_1 + y_2}{2}\right)$.

If line L has a slope of m and line K is perpendicular to line L, then line K has a slope of $-\dfrac{1}{m}$.

Practice problems

1. Simplify each expression

   (a) $3(x - 2) - 5(x + 3)$

   (b) $6(a - b) - (a + b)$

   (c) $3 - (-2)(-3)^2$

   (d) $(2 - 6)^2 \div (-2 - 6) + 4(-2)$

2. Translate into a mathematical expression with a variable

   (a) The quotient of a number and the difference between the number and 6

   (b) The sum of three times a number and the product of two and three less than the number

   (c) The product a number and of three more than the sum of the number and two

   (d) The sum of the square of twice a number and three times the number

3. Solve for the unknown

(a) $3a + 2(4a + 5) = 32$

(b) $5(x - 1) = 19 - (x + 2)$

(c) $\dfrac{3 + n}{12} = \dfrac{5}{6}$

(d) $\dfrac{s - 2}{3} + \dfrac{s}{5} = \dfrac{-2}{15}$

4. Is the point $(2,\ -3)$ on the line $2y + 2 + 6x = 0$? Show your work.

5. Find the equations of the following lines.

(a) The line throught $(0,\ 5)$ with slope 3.

(b) The line through $(2,\ -3)$ with slope $\dfrac{2}{5}$.

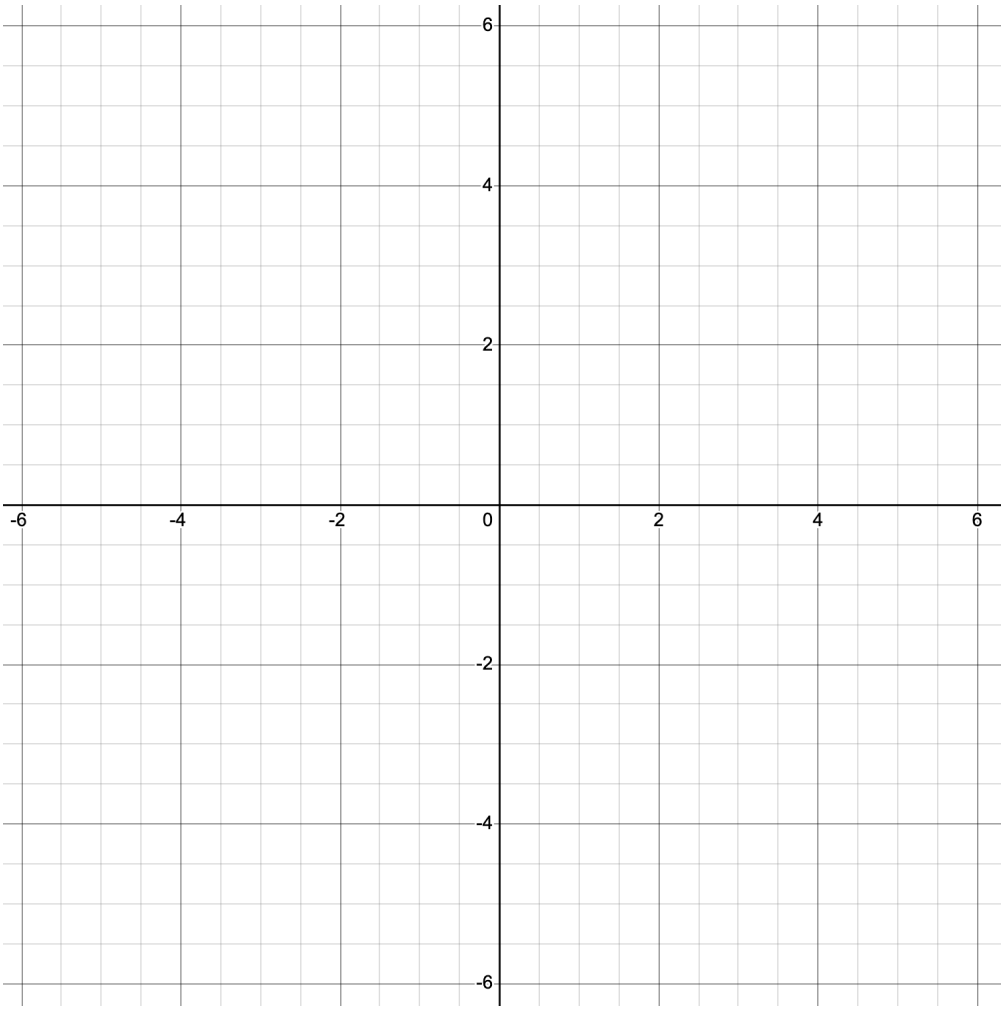
6. Given the points $P(3,\ -2)$ and $Q(5,\ 7)$:

(a) Find the coordinates of the midpoint of the line segment PQ.

(b) Find the slope of the line through P and Q.

(c) Find the equation of the perpendicular bisector of the line segment PQ.

7. Graph the lines $y = \dfrac{3}{4}x - 2$ and $2y + 3x - 4 = 0$ on the coordinate plane below.

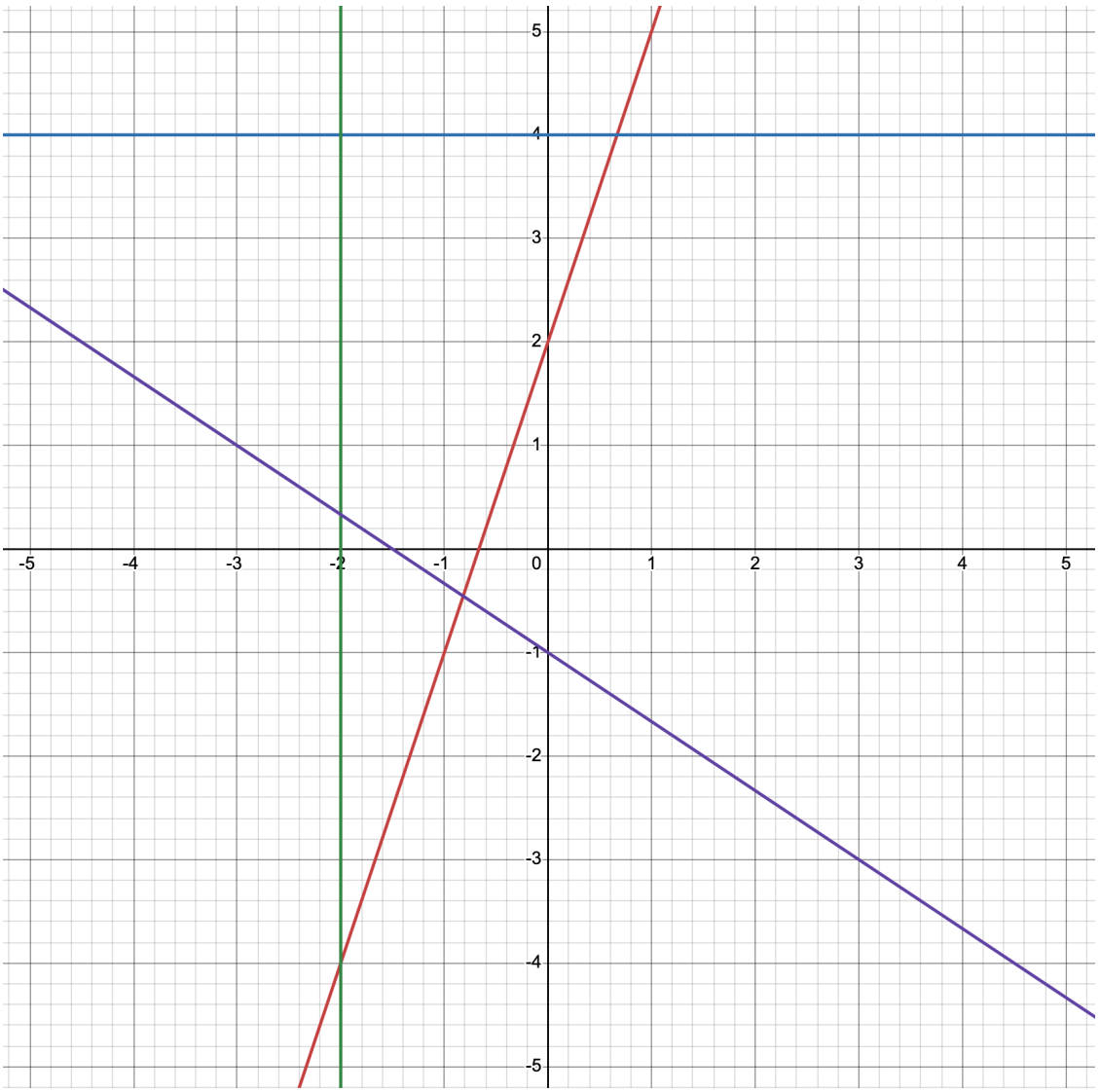


8. Find the slope of the lines through the points:

(a) $A(-2,\ 7), B(3,\ -3)$

(b) $A(0,\ 5), B(2,\ 7)$

9. Write down the equation of each of the four lines shown below.


5
4
3
2
1
-5
-4
-3
-2
-1
0
1
2
3
4
5
-1
-2
-3
-4
-5